Synoptic Question 14: The Physics of a Bungee Jump

Reading Passage

Read the following article and then answer the questions that follow.

A bungee jump is an extreme sport that involves jumping from a tall structure while connected to a long elastic cord. The thrill of the jump comes from the initial period of free fall, followed by a rapid deceleration as the cord stretches and slows the jumper, and a series of rebounds.

The cord itself is a specially designed elastic rope, typically made of many strands of latex rubber. For safety, the cord is designed to have a specific unstretched length (L_0) and a force constant (k) that are appropriate for the jumper's mass and the height of the platform. The cord must be strong enough to withstand the maximum tension experienced during the jump, which occurs when the jumper reaches the lowest point.

The jump can be analysed in several distinct phases: 1. **Free Fall:** From the platform, the jumper falls a distance equal to the cord's unstretched length. 2. **Stretching:** The cord begins to stretch, and its tension force increases, obeying Hooke's Law. The jumper continues to accelerate downwards, but at a decreasing rate. 3. **Equilibrium Point:** The jumper reaches a point where the upward tension force exactly equals their weight. The jumper is still moving downwards but is no longer accelerating. 4. **Deceleration:** Below the equilibrium point, the tension force exceeds the jumper's weight, causing a net upward force that decelerates the jumper, slowing them to a stop at the lowest point. 5. **Rebound:** The stored elastic potential energy in the cord pulls the jumper back upwards, initiating a series of oscillations that are damped (lose energy) over time.

Data:

A bungee jump is set up from a platform above a river.

Table 1: Data for Bungee Jump

Parameter	**Value**
Mass of the jumper (with harness), m	70.0 kg
Height of platform above river, H	110 m
Unstretched length of bungee cord, L_0	30.0 m
Stiffness (force) constant of the cord, k	65.0 N m^{-1}
Acceleration of free fall, g	9.81 m s^{-2}

Questions

(a) The jumper steps off the platform and is in free fall.

  (i) State the jumper's gravitational potential energy (GPE) relative to the river just before jumping.

  (ii) Calculate the jumper's speed, v_1, at the point where the bungee cord just begins to stretch. (Assume air resistance is negligible).

(b) The cord continues to stretch until the jumper momentarily stops at the lowest point of the jump. Let this maximum extension be x_{max}.

  (i) State the total vertical distance D fallen by the jumper from the platform in terms of L_0 and x_{max}.

  (ii) By considering the conservation of energy (GPE loss and EPE gain) from the platform to the lowest point, show that the maximum extension x_{max} satisfies the equation:

$$\frac{1}{2}kx_{max}^2 - mgx_{max} - mgL_0 = 0$$

  (iii) Calculate the value of the maximum extension, x_{max}.

(c) Using your value for x_{max}, calculate:

(i) The maximum tension T_{max} in the bungee cord.

(ii) The net force on the jumper at the lowest point of the jump.

(iii) The maximum acceleration experienced by the jumper.

(d) After reaching the lowest point, the jumper oscillates up and down around an equilibrium position.

(i) Calculate the extension x_e of the cord when the jumper hangs at rest (the equilibrium position).

(ii) Calculate the period of these oscillations, assuming they are Simple Harmonic Motion (SHM).

(iii) The amplitude A of the first oscillation is the distance from the equilibrium position to the lowest point. Calculate this amplitude.

(iv) Using your answers from (d)(ii) and (d)(iii), calculate the maximum speed of the jumper as they oscillate.

(e) The graph in Fig. 14.2 shows a conceptual sketch of the jumper's oscillations.

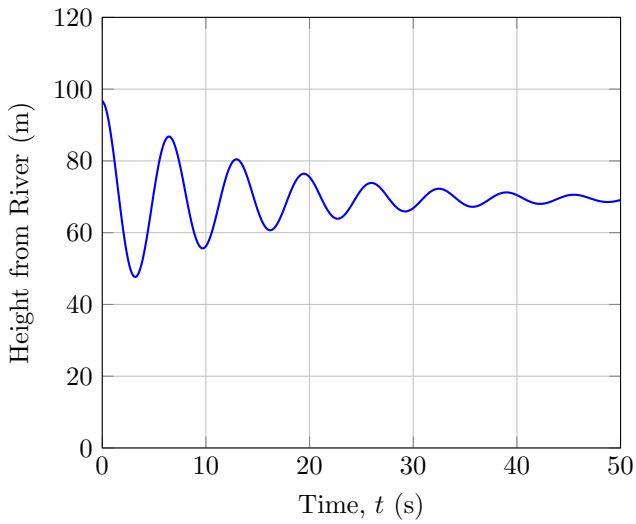

Figure 1: Conceptual graph of jumper's height over time

(i) What is the name given to this type of oscillation where the amplitude decreases over time?

(ii) State the main non-conservative force responsible for this energy loss.

(iii) The total energy of the oscillation is proportional to the square of its amplitude ($E \propto A^2$). Calculate the percentage of the oscillation energy that has been lost when the amplitude has decreased to 25% of its initial value.

Solutions

(a) (i) $GPE = mgH = (70.0)(9.81)(110) = 75537 \text{ J} \approx 7.55 \times 10^4 \text{ J}$.

(ii) The jumper falls a distance $L_0 = 30.0$ m.
Loss in GPE = Gain in KE
$mgL_0 = \frac{1}{2}mv_1^2$
$v_1 = \sqrt{2gL_0} = \sqrt{2(9.81)(30.0)} = 24.26 \text{ m s}^{-1} \approx 24.3 \text{ m s}^{-1}$.

(b) (i) $D = L_0 + x_{max}$.

(ii) Let's use the platform as the $GPE = 0$ reference point.
GPE at top = 0. KE at top = 0. EPE at top = 0. Total Energy = 0.
At the lowest point (distance D below platform):
$GPE_{bottom} = -mgD = -mg(L_0 + x_{max})$
$KE_{bottom} = 0$
$EPE_{bottom} = \frac{1}{2}kx_{max}^2$
By conservation of energy: $E_{top} = E_{bottom}$
$0 = -mg(L_0 + x_{max}) + \frac{1}{2}kx_{max}^2$
$0 = -mgL_0 - mgx_{max} + \frac{1}{2}kx_{max}^2$
$\frac{1}{2}kx_{max}^2 - mgx_{max} - mgL_0 = 0$. (Proven)

(iii) Substitute values into the quadratic equation for x_{max}:
$\frac{1}{2}(65.0)x_{max}^2 - (70.0)(9.81)x_{max} - (70.0)(9.81)(30.0) = 0$
$32.5x_{max}^2 - 686.7x_{max} - 20601 = 0$
Using $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$:
$x_{max} = \frac{686.7 \pm \sqrt{(-686.7)^2 - 4(32.5)(-20601)}}{2(32.5)}$
$x_{max} = \frac{686.7 \pm \sqrt{471557 + 2678130}}{65.0} = \frac{686.7 \pm 1774.7}{65.0}$
Since x_{max} must be positive, $x_{max} = \frac{2461.4}{65.0} = 37.87 \text{ m} \approx 37.9 \text{ m}$.

(c) (i) $T_{max} = kx_{max} = (65.0)(37.87) = 2461.6 \text{ N} \approx 2460 \text{ N}$.

(ii) Weight $W = mg = (70.0)(9.81) = 686.7$ N.
F_{net} (upwards) $= T_{max} - W = 2461.6 - 686.7 = 1774.9 \text{ N} \approx 1770 \text{ N}$.

(iii) $F_{net} = ma_{max}$
$a_{max} = \frac{F_{net}}{m} = \frac{1774.9}{70.0} = 25.36 \text{ m s}^{-2}$.

(d) (i) At equilibrium, the upward tension equals the downward weight.
$F_{tension} = W \implies kx_e = mg$
$x_e = \frac{mg}{k} = \frac{(70.0)(9.81)}{65.0} = 10.56$ m.

(ii) For a mass-spring system, $T = 2\pi\sqrt{\frac{m}{k}}$.
$T = 2\pi\sqrt{\frac{70.0}{65.0}} = 6.51$ s.

(iii) The lowest point is at extension $x_{max} = 37.87$ m. The equilibrium position is at extension $x_e = 10.56$ m.
The amplitude A is the maximum displacement from equilibrium.
$A = x_{max} - x_e = 37.87 - 10.56 = 27.31 \text{ m} \approx 27.3$ m.

(iv) Maximum speed in SHM is $v_{max} = \omega A$.
$v_{max} = \left(\frac{2\pi}{T}\right)A = \left(\frac{2\pi}{6.51}\right)(27.31) = 26.3 \text{ m s}^{-1}$.

(e) (i) This motion is called **damped oscillations** (or light damping).

(ii) **Air resistance** (or air drag) acting on the jumper.

(iii) $E_{initial} \propto A_{initial}^2$.
$E_{final} \propto (0.25A_{initial})^2 = 0.0625A_{initial}^2$.
The final energy is $E_{final} = 0.0625E_{initial}$, or 6.25% of the initial energy.
Percentage lost = 100% − 6.25% = 93.75%.

CIE 9702 Physics Paper 4 (A Level) – Variant 2 (Higher Difficulty)

Topic: Bungee Jump Dynamics, Energy Transfer and Modelling

Data for this variant

- Mass of jumper (with harness): $m = 74.0\,\text{kg}$
- Platform height above river: $H = 120\,\text{m}$
- Unstretched cord length: $L_0 = 32.0\,\text{m}$
- Cord stiffness constant: $k = 72.0\,\text{Nm}^{-1}$
- Gravitational field strength: $g = 9.81\,\text{ms}^{-2}$

Additional modelling assumptions (unless stated otherwise):

- Air resistance during the first descent is negligible.
- The cord obeys Hooke's law throughout extension.
- Cord mass is negligible.

Question 1

The jumper steps off the platform from rest and falls until the cord just becomes taut.

(a) Calculate the speed v_1 when the cord just begins to stretch.

(b) Determine the kinetic energy of the jumper at this instant.

(c) Explain, with reference to force and acceleration, why the speed continues to increase for a short interval even after the cord starts stretching.

Question 2

Let x be the extension of the cord beyond L_0, and let x_{max} be the extension at the lowest point.

(a) Show that x_{max} satisfies

$$\frac{1}{2}kx_{\text{max}}^2 - mgx_{\text{max}} - mgL_0 = 0.$$

(b) Calculate x_{max}.

(c) Hence determine the minimum clearance above the river at the lowest point, and comment on safety.

Question 3

At the lowest point, the cord tension is maximum.

(a) Calculate the maximum tension $T_{\max}$ in the cord.

(b) Calculate the maximum acceleration magnitude in units of g.

(c) A safety code requires peak deceleration not to exceed $3.5g$. State whether this setup satisfies the code.

Question 4

After several oscillations, the jumper is approximately in vertical SHM about an equilibrium extension x_e.

(a) Determine x_e.

(b) Calculate the period T of small oscillations.

(c) Determine the amplitude of the first upward rebound measured from equilibrium, assuming negligible damping over the first cycle.

(d) Find the maximum speed during this first cycle.

Question 5 (Open-ended physics analysis)

In practice, the jumper experiences significant non-conservative effects and the cord can show non-ideal elastic behaviour.

(a) Discuss two limitations of the simple model used above and explain qualitatively how each limitation would affect:

- the predicted lowest point,
- the peak tension,
- and the oscillation period.

(b) Design an experiment (measurements, apparatus and analysis method) to estimate an effective damping coefficient for the oscillation of a jumper-cord system.

(c) Explain how uncertainty in jumper mass and cord stiffness should be propagated to establish a conservative safe operating envelope for commercial bungee jumps.

Sample Solution (Variant 2)

Question 1

(a)
$$v_1 = \sqrt{2gL_0} = \sqrt{2(9.81)(32.0)} = 25.1\,\mathrm{ms}^{-1}.$$

(b)
$$E_k = \tfrac{1}{2}mv_1^2 = \tfrac{1}{2}(74.0)(25.1)^2 \approx 2.33 \times 10^4\,\mathrm{J}.$$

(c) Just after the cord becomes taut, extension x is small so $T = kx$ is initially less than mg. Resultant force is still downward, so acceleration remains downward and speed still increases. Speed reaches maximum when $T = mg$, after which acceleration becomes upward.

Question 2

(a) From platform to lowest point:
$$mg(L_0 + x_{\max}) = \tfrac{1}{2}kx_{\max}^2,$$
rearranging gives
$$\tfrac{1}{2}kx_{\max}^2 - mgx_{\max} - mgL_0 = 0.$$

(b) Substitute values:
$$36x_{\max}^2 - 725.94x_{\max} - 23230.08 = 0.$$
Positive root gives
$$x_{\max} = 35.7\,\mathrm{m}\ (3\,\mathrm{s.f.}).$$

(c) Total drop $D = L_0 + x_{\max} = 32.0 + 35.7 = 67.7\,\mathrm{m}$. Clearance above river $= H - D = 120 - 67.7 = 52.3\,\mathrm{m}$. Therefore the jumper remains well above the river for these parameters.

Question 3

(a)
$$T_{\max} = kx_{\max} = 72.0(35.7) = 2.57 \times 10^3\,\mathrm{N}.$$

(b)
$$a_{\max} = \frac{T_{\max} - mg}{m} = \frac{2570 - 725.94}{74.0} = 24.9\,\mathrm{ms}^{-2}.$$
In g units:
$$\frac{a_{\max}}{g} = \frac{24.9}{9.81} = 2.54g.$$

(c) Since $2.54g < 3.5g$, the setup satisfies the stated deceleration code.

Question 4

(a) Equilibrium: $kx_e = mg$,

$$x_e = \frac{mg}{k} = \frac{725.94}{72.0} = 10.1\,\text{m}.$$

(b)

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{74.0}{72.0}} = 6.38\,\text{s}.$$

(c) First-cycle amplitude from equilibrium:

$$A = x_{\max} - x_e = 35.7 - 10.1 = 25.6\,\text{m}.$$

(d)

$$v_{\max} = \omega A = \frac{2\pi}{T}A = \frac{2\pi}{6.38}(25.6) = 25.2\,\text{ms}^{-1}.$$

Question 5 (Open-ended physics analysis) – indicative points

(a) Any two well-developed limitations, e.g.:

- Air resistance (speed-dependent drag) dissipates energy, usually reducing actual lowest-point drop and reducing subsequent amplitudes.
- Cord hysteresis/non-linear elasticity means loading and unloading curves differ; peak tension and rebound predictions from Hooke's law become inaccurate.
- Finite cord mass modifies effective oscillating mass and mode shape; period differs from $2\pi\sqrt{m/k}$.

(b) Experimental design should include: time-resolved position data (video/motion sensor), repeated trials, extraction of amplitude envelope $A(t)$, logarithmic decrement or exponential fit $A = A_0 e^{-\beta t}$, and inference of damping coefficient with uncertainty.

(c) Safe-envelope method should include: worst-case parameter bounds ($m_{\max}$, $k_{\min}$), uncertainty propagation (analytic partial derivatives or Monte Carlo), conservative clearance margin, and acceptance criterion based on upper-bound drop and tension.

Mark Scheme (CIE 9702 P4 style) – Variant 2 only

Total: 40 marks

Question 1 [6 marks]

- (a) Uses $v = \sqrt{2gL_0}$ correctly [1]
- Correct substitution and $v_1 = 25.1\,\mathrm{ms}^{-1}$ (allow 3 s.f.) [1]
- (b) Uses $E_k = \frac{1}{2}mv^2$ [1]
- Correct $E_k \approx 2.33 \times 10^4$ J [1]
- (c) States $T < mg$ initially after tautening so resultant force still downward [1]
- Correctly identifies turning point condition related to $T = mg$ / acceleration sign change [1]

Question 2 [10 marks]

- (a) Clear energy statement from top to lowest point [1]
- Correct equation $mg(L_0 + x_{\max}) = \frac{1}{2}kx_{\max}^2$ [1]
- Rearranged shown form correct [1]
- (b) Correct substitution into quadratic [1]
- Valid method solving quadratic [1]
- Selects physically valid positive root [1]
- $x_{\max} = 35.7\,\mathrm{m}$ (allow 35.6–35.8) [1]
- (c) Computes $D = L_0 + x_{\max}$ [1]
- Computes clearance $H - D = 52.3\,\mathrm{m}$ [1]
- Valid safety comment linked to positive clearance [1]

Question 3 [8 marks]

- (a) Uses $T_{\max} = kx_{\max}$ [1]
- Correct $T_{\max} \approx 2.57 \times 10^3$ N [1]
- (b) Uses resultant force $T_{\max} - mg$ [1]
- Correct acceleration in ms^{-2} [1]
- Correct conversion to $2.54g$ (or equivalent) [1]
- (c) Compares with $3.5g$ limit explicitly [1]
- Correct decision (complies) [1]
- Appropriate unit/significant figures throughout [1]

Question 4 [8 marks]

- (a) Uses $x_e = mg/k$ [1]
- Correct $x_e = 10.1\,\mathrm{m}$ [1]
- (b) Uses $T = 2\pi\sqrt{m/k}$ [1]
- Correct $T = 6.38\,\mathrm{s}$ [1]
- (c) Uses $A = x_{\max} - x_e$ [1]
- Correct $A = 25.6\,\mathrm{m}$ [1]
- (d) Uses $v_{\max} = \omega A$ or $2\pi A/T$ [1]
- Correct $v_{\max} = 25.2\,\mathrm{ms}^{-1}$ [1]

Question 5 [8 marks]

- (a) Two relevant model limitations identified [2]
- Physics reasoning on effects for drop/tension/period is coherent and specific [2]
- (b) Experimental method includes measurable variables, apparatus and analysis route [2]
- (c) Uncertainty propagation and conservative safety-envelope logic clearly explained [2]

Examiner note: For Question 5, award credit for any physically valid, well-argued response consistent with A Level mechanics and oscillations.